

STA 35C – Homework 7

Submission due: Tue, June 2 at 11:59 PM PT

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Instructions: Upload a **single PDF file** to Gradescope via Canvas (“Homework 7” under “Assignments”). Name it using your UC Davis email ID prefix and homework number, e.g., `dgsong_hw7.pdf`. Include “STA 35C,” your name, and the last four digits of your student ID on the front page. For coding problems, include your R code, relevant output, and requested plots/tables in the PDF. Late submissions receive 0 points; see the syllabus for full submission and late-policy details.

Problem 1: Principal component analysis (40 points total).

You have a *two-dimensional* dataset of 10 points, $\{x_1, \dots, x_{10}\} \subset \mathbb{R}^2$. Suppose the 10 data points are:

$$\begin{aligned} x_1 &= (2, 3), & x_2 &= (2, 5), & x_3 &= (3, 6), & x_4 &= (4, 5), & x_5 &= (5, 8), \\ x_6 &= (6, 10), & x_7 &= (6, 7), & x_8 &= (7, 11), & x_9 &= (7, 9), & x_{10} &= (8, 12). \end{aligned}$$

(a) (10 points)

- (i) Compute the sample mean $\bar{x} \in \mathbb{R}^2$ and *center* the data by subtracting $\bar{x}$ from each point.
- (ii) Compute the 2×2 *sample covariance matrix*:

$$\Sigma = \frac{1}{10} \sum_{i=1}^{10} (x_i - \bar{x})(x_i - \bar{x})^\top.$$

(Note: Typically, we treat each data point x_i as a (column) vector, e.g., $x_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Then its transpose

$x_i^\top$ is the row vector, e.g., $x_1^\top = [2 \ 3]$. You can compute the product $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \begin{bmatrix} b_1 & b_2 \end{bmatrix} = \begin{bmatrix} a_1 b_1 & a_1 b_2 \\ a_2 b_1 & a_2 b_2 \end{bmatrix}$.

In **R**, you can compute such an outer product by `a %*% t(b)`, where **a** and **b** are column vectors.)

- (iii) Using **R** (e.g. `prcomp` or an eigen-decomposition of Σ), find the *first principal component direction* $\mathbf{u}_1$. (Note: The sign of $\mathbf{u}_1$ as a principal component is arbitrary, so $\mathbf{u}_1$ and $-\mathbf{u}_1$ are equivalent.)

(b) (10 points)

- (i) Compute the *directional variance* along $\mathbf{e}_1 = (1, 0)$; i.e. the variance of $\langle \mathbf{e}_1, x_i \rangle$.
 - (ii) Compute the directional variance along your first principal component $\mathbf{u}_1$ from part (a)-(iii).
 - (iii) Verify that these directional variances match $\mathbf{e}_1^\top \Sigma \mathbf{e}_1$ and $\mathbf{u}_1^\top \Sigma \mathbf{u}_1$ for Σ computed above, respectively.
- (c) (10 points) In **R**, generate a *synthetic* dataset with $p = 25$ variables and $n = 100$ observations. For example, use the following code snippet (mixture of 4 Gaussians, with two means close together):

```

set.seed(2026)
n  <- 100
p  <- 25

mean1 <- c(2, rep(0, p-1))
mean2 <- c(0, 2, rep(0, p-2))
mean3 <- c(1, 1.5, -1, rep(0, p-3))
mean4 <- c(1, 1, 1, rep(0, p-3))

X1 <- MASS::mvrnorm(n, mean1, diag(p))
X2 <- MASS::mvrnorm(n, mean2, diag(p))
X3 <- MASS::mvrnorm(n, mean3, diag(p))
X4 <- MASS::mvrnorm(n, mean4, diag(p))

# Combine a "mixture" of 30 + 30 + 20 + 20 = 100 obs from 4 subgroups
X <- rbind(X1[1:30,], X2[1:30,], X3[1:20,], X4[1:20,])

```

- (i) Identify the first principal component direction and compute the directional variance along it.
 - (ii) Run PCA (e.g. using `prcomp`) and create a *scree plot* showing the proportion of variance explained by each principal component.
- (d) (10 points)** Using the synthetic 25-dimensional dataset from part (c), compare the following two clustering approaches.
- (i) Run *k-means* with $k = 4$ on the original 25-dimensional data. Visualize the resulting clusters on the PC1–PC2 plane.
 - (ii) Project the data onto the first two principal components (PC1 and PC2), then run *k-means* clustering with $k = 4$ on these two-dimensional PC scores.

For each approach, draw a scatter plot on the PC1–PC2 plane, coloring points by the assigned cluster. Then briefly discuss how PCA can help (or might not help) before applying k-means in a high-dimensional setting. Thereafter, for a brief sensitivity check, repeat the PC-score k-means analysis with $k = 3$ and $k = 5$, and comment in one or two sentences.

Problem 2: K-means clustering (35 points total).

You have $n = 10$ data points, each representing a local retailer's attributes. Two features are:

- $x = \text{store floor area}$ (in hundreds of m^2),
- $y = \text{annual revenue}$ (in thousands of \$).

Below are the measurements $\{z_i = (x_i, y_i)\}_{i=1}^{10}$:

$$\begin{array}{llllll}
 z_1 = (2, 4), & z_2 = (3, 3), & z_3 = (3, 6), & z_4 = (5, 15), & z_5 = (6, 12), \\
 z_6 = (4, 2), & z_7 = (10, 22), & z_8 = (11, 20), & z_9 = (12, 25), & z_{10} = (8, 16).
 \end{array}$$

We want to form $K = 3$ clusters using *k-means* clustering.

- (a) (10 points) **Explain k-means clustering** in your own words, including:
- What is the objective function it aims to minimize?
 - How does the algorithm alternate between assigning points to clusters and updating centroids?
 - Why do we need to pick K in advance, and how might we choose K in practice?
- (b) (5 points) Draw a scatter plot of $z_i = (x_i, y_i)$. Briefly describe any obvious grouping you see (if any).
- (c) (10 points) **Perform two iterations of k-means** using $K = 3$ clusters on the data $\{z_1, \dots, z_{10}\}$ above. Start with an initial assignment:

$$C_1 = \{z_1, z_6\}, \quad C_2 = \{z_2, z_3, z_5\}, \quad C_3 = \{z_4, z_7, z_8, z_9, z_{10}\}.$$

Carry out:

- (i) Compute centroids of C_1, C_2, C_3 .
- (ii) Reassign each z_i to the closest centroid.
- (iii) Recompute centroids, then reassign again.

Show your arithmetic through hand calculation or R (stop after 2 full updates).

- (d) (10 points) **Try random initialization:** If you randomly assign points to 3 clusters at the start, does the final solution differ? Repeat k-means with 5 different random starts, and note any differences in the final cluster membership or total within-cluster sum of squares.

Problem 3: Hierarchical clustering (25 points total).

- (a) (5 points) In your own words, explain the hierarchical clustering algorithm, including:
- How does it start, and how are “closest” clusters merged at each step?
 - List at least two commonly used distance measures between clusters.
- (b) (10 points) [JWHT21, Chapter 12, Exercise 4].
- (c) (10 points) Consider the dataset from Problem 2. Use hierarchical clustering with **complete linkage** and Euclidean distance on the raw coordinates. Show the successive merges, specifically answering:
- Which points/clusters merge first?
 - Is there a noticeable large jump in merge height? If so, where does it occur?
 - About how many merges occur before a “major” grouping forms?

Sketch or print a dendrogram. Identify a horizontal cut that yields 3 clusters, and compare these clusters to the k-means results from Problem 2.

Problem 4: Bonus problems (10 bonus points).

- (a*) (*5 bonus points) [JWHT21, Chapter 12, Exercise 5].
- (b*) (*5 bonus points) [JWHT21, Chapter 12, Exercise 6].

References

- [JWHT21] Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An Introduction to Statistical Learning: with Applications in R*, volume 112 of *Springer Texts in Statistics*. Springer, New York, NY, 2nd edition, 2021.